



# VIT<sup>®</sup>

Vellore Institute of Technology  
(Deemed to be University under section 3 of UGC Act, 1956)  
CHENNAI

23MCE21460

## Final Assessment Test(FAT) - NOV/DEC 2025

|              |                         |              |                         |
|--------------|-------------------------|--------------|-------------------------|
| Programme    | B.Tech.                 | Semester     | Fall Semester 2025-26   |
| Course Code  | BCSE306L                | Faculty Name | Prof. Modigari Narendra |
| Course Title | Artificial Intelligence | Slot         | D1+TD1                  |
|              |                         | Class Nbr    | CH2025260100594         |
| Time         | 3 hours                 | Max. Marks   | 100                     |

### Instructions To Candidates

- Write only your registration number in the designated box on the question paper. Writing anything elsewhere on the question paper will be considered a violation.

### Course Outcomes

- CO1: Evaluate Artificial Intelligence (AI) methods and describe their foundations.  
CO2: Apply basic principles of AI in solutions that require problem-solving, inference, perception, knowledge representation and learning.  
CO3: Demonstrate knowledge of reasoning, uncertainty, and knowledge representation for solving real-world problems  
CO4: Analyse and illustrate how search algorithms play a vital role in problem-solving

### Section - I

#### Answer all Questions (3 × 12 Marks)

01. An autonomous car needs to drive from Home (S) to Work (G) in a city with roads of varying traffic congestion levels. Each road segment has a travel cost (time in minutes), and a navigation model predicts estimated remaining time to the destination based on GPS data. The available road connections and travel times are given below:

| From | To | Travel Time (minutes) |
|------|----|-----------------------|
| S    | A  | 4                     |
| S    | B  | 6                     |
| A    | C  | 5                     |
| A    | D  | 11                    |
| B    | C  | 3                     |
| B    | D  | 8                     |
| C    | E  | 6                     |
| D    | G  | 5                     |
| E    | D  | 3                     |
| E    | G  | 4                     |

The estimated remaining time to the goal (G) is:

| Node to G | Estimated Time (minutes) |
|-----------|--------------------------|
| S         | 10                       |
| A         | 8                        |
| B         | 7                        |
| C         | 6                        |
| D         | 3                        |
| E         | 2                        |
| G         | 0                        |

- a. Using an informed search strategy that combines both actual travel time so far and the predicted remaining time to the destination, determine the optimal route from S to G. At each step, clearly show: (8 marks)
- The node selected for expansion, and calculated values of  $g(n)$ ,  $h(n)$ , and  $f(n)$
  - The contents of the open list and the closed list
  - The search tree is explored during the route calculation process, with each node labelled by its  $f(n)$  value.

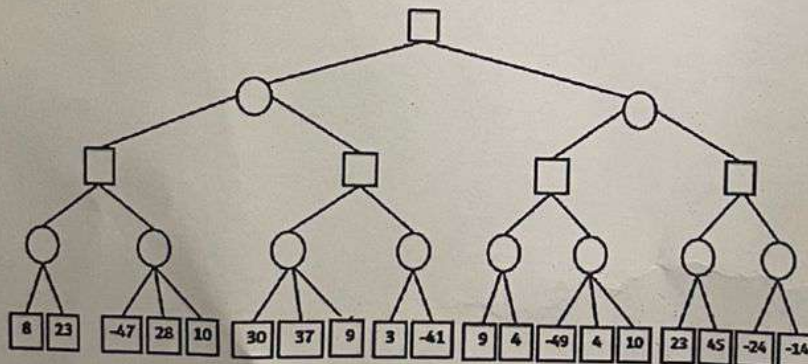
- iv. Mark the chosen path from S to G and specify the total travel time for the optimal route.
- b. If the predictive model were unavailable (i.e., all heuristic values were set to zero), what traditional search strategy would the car's navigation system effectively perform? Would this method still guarantee the optimal path? Explain briefly. (4 marks)

[12] (CO<sub>4</sub>/K<sub>3</sub>)

02. A logistics company is developing an AI-based autonomous delivery system where drones pick up parcels, navigate urban areas, avoid obstacles, and deliver packages to customers' doorsteps. The system learns from past routes, predicts weather-related risks, interacts with other drones and human controllers, and continuously improves delivery efficiency.
- Compare the current type of AI used in this system with a more advanced type that could improve its decision-making. What additional capabilities or components must be incorporated to achieve that advancement? (3 Marks)
  - Formulate the complete PEAS (Performance measure, Environment, Actuators, Sensors) description for this AI agent. (4 Marks)
  - Classify and explain the nature of the environment in which this agent operates with respect to the following characteristics: (5 Marks)
    - Observability
    - Determinism
    - Episodicity
    - Static or Dynamic nature
    - Number of agents

[12] (CO1/K2)

03. Imagine you are designing an AI agent to play a simplified version of a chess game with the game tree as shown below. The root node represents Player\_White (Max), and the levels alternate between Player\_White (Max) and Player\_Black (Min). The leaf nodes display static evaluation scores of the board positions from Player\_White's perspective.



- Apply an adversarial search technique to identify the moves that Player\_White at the root should select to achieve the highest possible static evaluation score. Explain the uninformed search strategy applied in this context. (4 Marks)
- Explain in detail the search technique used to determine the optimal path for Player\_White to maximize the static evaluation score while minimizing the exploration of unnecessary nodes. (6 Marks)
- Compare the performance of both search techniques applied in (a) and (b) (2 Marks)

[12] (CO4/K3)

## Section - II

**Answer all Questions (4 × 16 Marks)**

04. A moving autonomous vending bot WERNA sells ice cream to students on the VIT campus. When a student scans the QR code for an item, WERNA attempts to receive payment and opens the corresponding lid. After the student picks the item, the lid closes. If the chosen item is sold out, WERNA substitutes vanilla (which is never out of stock). If the student's account lacks sufficient balance, payment fails and the order is canceled.
- (a) Shreyas a student has enough money to buy either butterscotch or vanilla, and scans the QR code for butterscotch, but butterscotch is unavailable. The goal is for WERNA to deliver an ice cream to Shreyas. Represent this scenario using a Planning Domain Description Language by choosing the objects, predicates, start state, goal state, and at least three actions. (8 Marks)
- (b) Let us consider WERNA has a navigation map as shown in Fig. 1. It is currently at LIBRARY and must reach AB3. Using the predicates from Fig. 1 and the action schema in Fig. 2, represent the start and goal states. Also perform Goal Stack Planning to produce the action sequence from start to goal. Show each step of the

algorithm, printing the stack contents and the current state every time an action or subgoal is pushed or popped and when an action is applied. (8 Marks)

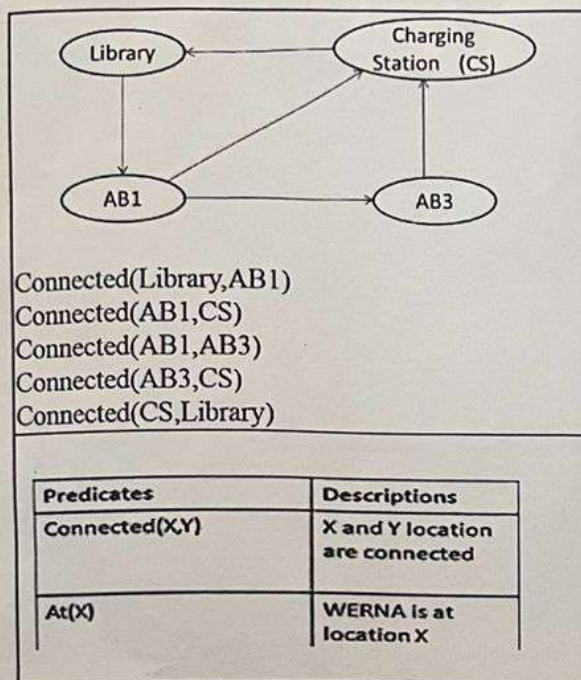


Fig 1. Map Given to WERNA and the Domain Predicates

| ActionName  | PreCondition            | Add Effects | DeleteEffects |
|-------------|-------------------------|-------------|---------------|
| Travel(X,Y) | At(X)<br>Connected(X,Y) | At(Y)       | ~At(X)        |

Fig2. Action Schema Defined for WERNA

[16] (CO2/K3)

05. The following statements describe the behavior of a group of students in a learning environment:

- All students learn at least one subject.
- Students who learn physiotherapy or osteopathy tend to avoid mathematics.
- Some students assume that anyone who delays any subject is imprudent.
- Anyone who avoids computation is imprudent.
- A person who delays any subject or avoids any computation is considered imprudent

Tasks:

- Represent all the above statements in First-Order Logic (FOL) clearly, defining appropriate predicates. (5 Marks)
- Convert the FOL statements step-by-step into Conjunctive Normal Form (CNF). (5 Marks)
- Using resolution refutation, prove that "Any student who learns physiotherapy and delays a subject is imprudent." (6 Marks)

[16] (CO2/K3)

06. Foot XYZ is a company formed by a team of three football fans aims to create an analytics dashboard that can showcase the prediction about the probability of a team's victory based on various performance-related factors. The team considers the training Intensity (T), player fitness (F), team morale (M), weather condition (W), and home ground (H) as key indicators to predict team's overall match performance (P). T impacts both F and M which together determines performance, P. Weather during the match affects F and M. In contrast, H impacts team morale, M and will affect strength of opponent strength (O) on the outcome. The final match outcome, Victory (V), depends primarily on the team's match performance and the strength of the opponent.

- Draw a Bayesian network for the above variables showing all dependencies. Identify the independent variables in the network. (3 Marks)
- Assume the following CPT values:

| Node | Parents (Given) | Values           | P(Node = High/Good/Win) | P(Node = Low/Poor/Loss) |
|------|-----------------|------------------|-------------------------|-------------------------|
| T    | —               | —                | 0.7                     | 0.3                     |
| W    | —               | —                | 0.6                     | 0.4                     |
| H    | —               | —                | 0.5                     | 0.5                     |
| F    | T,W             | High, Good       | 0.9                     | 0.1                     |
| F    | T,W             | High, Bad        | 0.7                     | 0.3                     |
| F    | T,W             | Low, Good        | 0.6                     | 0.4                     |
| F    | T,W             | Low, Bad         | 0.3                     | 0.7                     |
| M    | T,W,H           | High, Good, Home | 0.95                    | 0.05                    |
| M    | T,W,H           | High, Good, Away | 0.85                    | 0.15                    |
| M    | T,W,H           | High, Bad, Home  | 0.8                     | 0.2                     |
| M    | T,W,H           | High, Bad, Away  | 0.65                    | 0.35                    |
| M    | T,W,H           | Low, Good, Home  | 0.7                     | 0.3                     |
| M    | T,W,H           | Low, Good, Away  | 0.5                     | 0.5                     |
| M    | T,W,H           | Low, Bad, Home   | 0.4                     | 0.6                     |
| M    | T,W,H           | Low, Bad, Away   | 0.2                     | 0.8                     |

Construct the Conditional Probability Table (CPT) for Team Morale (M) for all combinations of T, W, and H. If T=High, W=Bad, H=Home, calculate the probability of M=High using the CPT provided. (3 Marks)

c. Assuming F=High and M=High, and that Team Performance (P), Compute the probability of Performance = Good. (3 Marks)

| F    | M    | P(P=Good) | P(P=Poor) |
|------|------|-----------|-----------|
| High | High | 0.95      | 0.05      |
| High | Low  | 0.7       | 0.3       |
| Low  | High | 0.6       | 0.4       |
| Low  | Low  | 0.2       | 0.8       |

d. Suppose the dashboard predicts a high probability of victory even when Opponent Strength (O)=Strong. Critically discuss two possible reasons using the relationships among T, F, M, P, and H. (4 Marks)

e. Suggest one method that could be used to estimate  $P(V=Win | T=High, W=Bad, H=Home)$  without calculating the full joint probability distribution. Briefly describe the steps involved. Also, comment on the usage of the technique over time complexity. (3 Marks)

[16] (CO3/K3)

07. A news recommendation system is designed to retrieve the most relevant articles when a user searches for "climate change policy and renewable energy". To evaluate relevance, the system applies the TF-IDF (Term Frequency-Inverse Document Frequency) algorithm to compute the importance of words in each article. Six simplified article snippets are given below:

- Document 1: "Climate change affects weather patterns and global temperature."
- Document 2: "Government introduces new climate policy to reduce emissions."
- Document 3: "Scientists discuss climate change and renewable energy policy."
- Document 4: "Sports policy committee reviews new training methods."
- Document 5: "Renewable energy projects receive funding to combat climate change."
- Document 6: "Economic analysts predict the impact of energy prices on renewable markets."

- a. Calculate the Term Frequency (TF) and Inverse Document Frequency (IDF) values for the following terms across all six documents: 'climate', 'change', 'policy', 'renewable', 'energy', 'government', 'emissions', and 'economic'. (Assume log base 10 for IDF and ignore stopwords.) (7 Marks)
- b. Using the values obtained in (a), calculate the TF-IDF scores for each of the given terms across the six documents. (5 Marks)
- c. Using the computed TF-IDF scores, determine which document is most relevant to the user query "climate change policy and renewable energy", and justify your answer based on the total TF-IDF weights for all query-related terms. (4 Marks)

[16] (CO1/K3)

BL-Bloom's Taxonomy Levels - (K1-Remembering, K2-Understanding, K3-Appling, K4-Analysing, K5-Evaluating, K6-Creating)

